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Studierichting: Informatica
Oefeningenreeks 4A nr. 37.4

$$\int \frac{5dx}{x^2 - 2x + 5}$$

$$= 5 \int \frac{dx \cdot \frac{1}{4}}{(x-1)^2 + 4}$$

$$= \frac{5}{4} \int \frac{dx}{\frac{1}{4}(x-1)^2 + 1}$$

$$= \frac{5}{4} \int \frac{dx}{\left(\frac{1}{2}(x-1)\right)^2 + 1}$$

Stel $u = \frac{1}{2}(x-1) \Rightarrow du = \frac{dx}{2} \Leftrightarrow dx = 2du$

$$= \frac{5}{2} \int \frac{du}{u^2 + 1}$$

$$= \frac{5}{2} \operatorname{Bgtan}(u) + c$$

$$= \frac{5}{2} \operatorname{Bgtan}\left(\frac{1}{2}(x-1)\right) + c$$

References